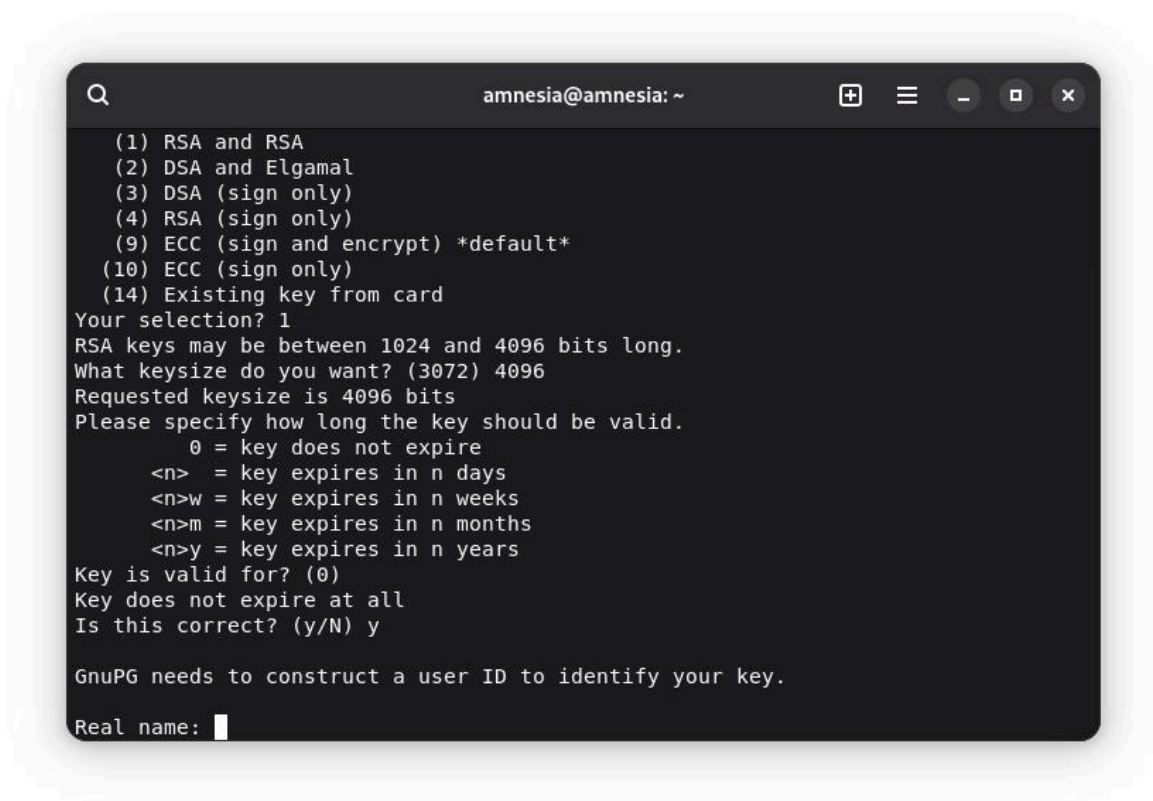
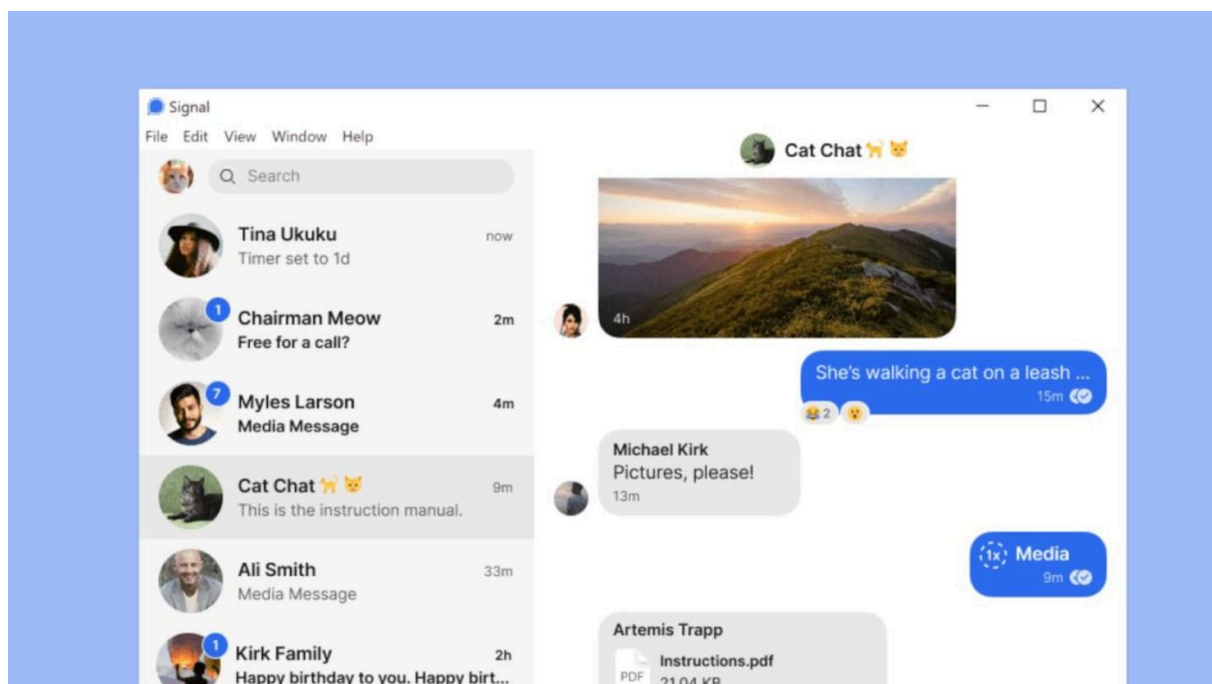
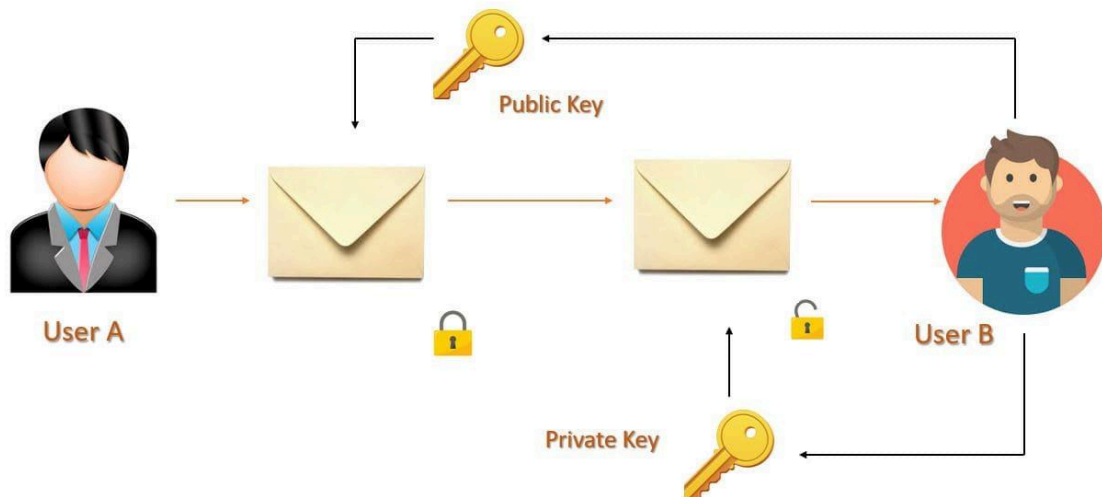


Secure Communications for Red Teams

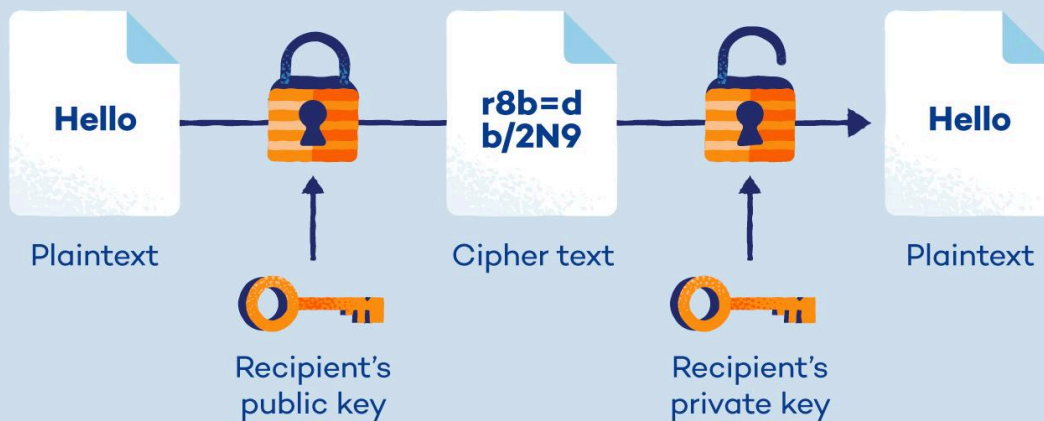
Secure communications are an important part of professional red team engagements because they help protect sensitive assessment data, maintain client confidentiality, and ensure authorized team members can communicate securely.

A screenshot of a terminal window titled 'amnesia@amnesia: ~'. The terminal displays the output of the 'gpg --full-genkey' command. It shows a list of key types: (1) RSA and RSA, (2) DSA and Elgamal, (3) DSA (sign only), (4) RSA (sign only), (9) ECC (sign and encrypt) *default*, (10) ECC (sign only), and (14) Existing key from card. The user selects '1'. The terminal then prompts for a key size, showing 'RSA keys may be between 1024 and 4096 bits long.' and 'What keysize do you want? (3072) 4096'. It then asks for the validity of the key, showing 'Please specify how long the key should be valid.' and options for days, weeks, months, and years. The user selects '0' for 'key does not expire'. The terminal then asks 'Key is valid for? (0)' and 'Key does not expire at all'. Finally, it asks 'Is this correct? (y/N) y' and 'GnuPG needs to construct a user ID to identify your key.' The prompt 'Real name: ' is visible at the bottom with a cursor.

How PGP Works?



PGP Encryption Process



STEP 1

PGP generates an unbreakable **random session key**.

STEP 2

The sender gets the **recipient's public key**, and uses that to encrypt and send the message.

STEP 3

The recipient gets the message, and uses their **unique private key** to decrypt the message and read it.

```

me@fedora ~-> gpg --gen-key
gpg (GnuPG) 2.3.4; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/me/.gnupg' created
gpg: keybox '/home/me/.gnupg/pubring.kbx' created
Note: Use "gpg --full-generate-key" for a full featured key generation dialog.

GnuPG needs to construct a user ID to identify your key.

Real name: example
Email address: example@mail.com
You selected this USER-ID:
    "example <example@mail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/me/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/me/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/me/.gnupg/openpgp-revocs.d/501B9825ED88ED009AD815734A82D15798B34A21.rev'
public and secret key created and signed.

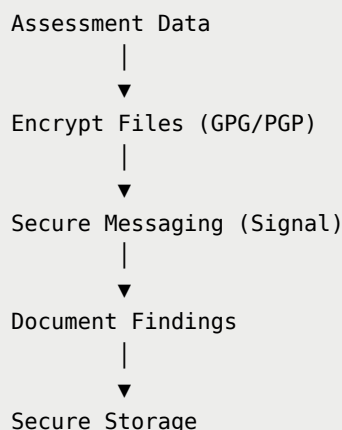
pub   ed25519 2022-07-30 [SC] [expires: 2024-07-29]
       501B9825ED88ED009AD815734A82D15798B34A21
uid           example <example@mail.com>
sub   cv25519 2022-07-30 [E] [expires: 2024-07-29]

me@fedora ~->

```

Secure Communications for Red Teams

A typical communication stack looks like this:



1. GPG / PGP

GNU Privacy Guard (GPG) is an implementation of the OpenPGP standard used for:

- Encrypting files
- Encrypting emails

- Digitally signing documents
- Verifying authenticity
- Sharing encrypted reports with clients

Common Uses

- Encrypting penetration test reports
- Protecting credentials during transfer
- Verifying that reports haven't been modified
- Securely sharing scripts or evidence

Core Concepts

Public Key

- Shared with others
- Used to encrypt data to you
- Used to verify your digital signature

Private Key

- Kept secret
 - Used to decrypt data
 - Used to create digital signatures
-

2. Digital Signatures

Digital signatures provide:

- Authentication (who sent it)
- Integrity (whether it changed)
- Non-repudiation (proof of origin)

This is useful when delivering assessment reports or scripts to clients.

3. Signal

Signal is widely used for secure team communication.

Features include:

- End-to-end encrypted messages
- Voice and video calls
- Group chats
- Disappearing messages
- Media sharing
- Cross-platform support

Common red team use cases include:

- Internal team coordination
 - Secure client communications (when agreed upon)
 - Sharing time-sensitive updates during engagements
-

4. Secure File Sharing

Rather than sending sensitive files in plain email, teams often:

- Encrypt reports with GPG
- Share passwords or decryption instructions through a separate communication channel
- Use approved encrypted collaboration platforms

This reduces the impact if one communication channel is compromised.

5. Identity Verification

Before exchanging sensitive information:

- Verify public key fingerprints through a trusted channel.
 - Confirm the identity of new contacts.
 - Rotate keys when appropriate.
 - Revoke compromised keys promptly.
-

6. Temporary ("Burner") Infrastructure Basics

In professional security work, the term **burner infrastructure** generally refers to **temporary, purpose-specific infrastructure** created for an engagement and then decommissioned afterward. This helps reduce long-term exposure and keeps client work separated.

Good operational practices include:

- Use separate infrastructure for each engagement.
- Avoid mixing client data across projects.
- Use dedicated domains, systems, and accounts for authorized testing.
- Remove or archive infrastructure securely after the engagement ends.
- Maintain clear asset inventories and logs.

These practices support good operational hygiene and client confidentiality.

Operational Security (OPSEC) Best Practices

- Encrypt sensitive files at rest and in transit.
 - Enable multi-factor authentication on collaboration platforms.
 - Use strong, unique passwords stored in a password manager.
 - Keep systems and software up to date.
 - Apply the principle of least privilege.
 - Separate personal and work accounts.
 - Back up encryption keys securely and maintain a revocation certificate.
-

Skills Worth Learning

If you're pursuing penetration testing or red teaming, these communication-related skills are valuable:

- OpenPGP/GPG key management

- Digital signatures and verification
 - Email encryption concepts
 - Secure messaging platforms
 - Public Key Infrastructure (PKI)
 - Certificate management
 - Operational security (OPSEC)
 - Secure report handling and evidence management
-

Official Resources

- [GnuPG Official Website](#)
- [Signal Official Website](#)
- [OpenPGP Standard \(RFC 9580\)](#)